exercise 9.

problem 1.

Sudoku can be modeled as a graph colouring problem.
Each cell of the 9x9 Sudoku grid is treated as a vertex.
So the graph has 81 vertices.
Two vertices are connected by an edge if the two cells are in:
① the same row, ② the same column, or ③ the same 3x3 subgrid.
The colors are the numbers 1 to 9.
Solving Sudoku means finding a proper 9-coloring of the graph, so that any two adjacent vertices have different colors.
This works because cells in the same row, column, or 3x3 box cannot have the same number.

Boundary conditions:

The given numbers in the sudoku puzzle are the boundary conditions (or fixed constraints).
If a cell is already filled with a number, then its color is fixed.
These fixed colors cannot be changed.
The remaining empty cells must be assigned colors that do not conflict with their adjacent vertices.
So, in graph terms, Sudoku is a graph coloring problem with some vertices already pre-colored.

problem 2.

Initialization: Set the source distance to 0, and all other distances to infinity. $d(S)=0$, $d(E)=\infty$, $d(D)=\infty$, $d(A)=\infty$
$d(C)=\infty$ $d(B)=\infty$

Iteration 1 one by one.

$S\to E=8$ $d(E)=8$. $S\to A=10$ $d(A)=10$, $E\to D=1$ $d(D)=8+1=9$
$D\to C=-1$ $d(C)=9-1=8$, $D\to A=-4$ $d(A)=\min(10, 9-4)=5$
$A\to C=2$ $d(C)=\min(8, 5+2)=7$, $A\to B=1$ $d(B)=5+1=6$
$C\to B=1$ $d(B)=\min(6, 7+1)=6$ no change.

So, after 1 Iteration:

$d(S)=0$, $d(E)=8$, $d(D)=9$, $d(A)=5$, $d(C)=7$, $d(B)=6$

Distance estimate after each iteration

There are 6 vertices, so Bellman-Ford performs $|V|-1=5$ iterations

Distance Table

Iteration	d(S)	d(E)	d(D)	d(A)	d(C)	d(B)
0 (initial)	0	∞	∞	∞	∞	∞
1	0	8	9	5	7	6
2	0	8	9	5	7	6
3	0	8	9	5	7	6
4	0	8	9	5	7	6
5	0	8	9	5	7	6

After the first iteration, all shortest distances are already found

From iteration 2 to iteration 5, no value changes.

Final shortest-path distances

The final shortest-path distances from S are:

$d(S)=0$, $d(E)=8$. $d(D)=9$, $d(A)=5$, $d(C)=7$ $d(B)=6$

Possible shortest paths:

$S\to E$: distance 8, $S\to E\to D$: distance 9

$S\to E\to D\to A$: distance 5, $S\to E\to D\to A\to C$: distance 7

$S\to E\to D\to A\to B$: distance 6.

Dijkstra's algorithm assumes that once a vertex gets the smallest temporary distance and is selected, its distance is final and will never become smaller later.

That assumption is not valid here because the graph has negative-weight edges:

$D\to C=-1$ $D\to A=-4$

For example

At first, we can get $A=10$ directly from $S\to A$.

But later, through $S\to E\to D\to A$, we get $8+1-4=5$ which is smaller than 10.

So a vertex that seems good earlier may still be improved later because of negative edge. Therefore, Dijkstra's greedy rule does not hold, and Dijkstra's algorithm is not reliable on this graph

5. Does the graph contain a negative cycle?

No, the graph does not contain a negative cycle

After the 5 iterations, no distance change anymore.

If we do one more check, no edge can still be relaxed.

Also, from the graph structure, there is actually no cycle at all.

So a negative cycle is impossible.